



Determining the etiology of adrenal insufficiency in adults

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INTRODUCTION

Adrenal insufficiency is broadly classified as primary, secondary, or tertiary based on whether cortisol deficiency derives from a defect in the adrenal glands, pituitary gland, or hypothalamus, respectively. Secondary and tertiary adrenal insufficiency are collectively referred to as central adrenal insufficiency. Once cortisol deficiency is confirmed, the next step is to determine the level of the defect leading to adrenal insufficiency. Establishing the etiology of adrenal insufficiency is important because it may be due to a disease that has other clinical ramifications, such as tuberculosis or an autoimmune polyglandular syndrome (in primary adrenal insufficiency) or a pituitary tumor (in secondary adrenal insufficiency). (See "[Causes of secondary and tertiary adrenal insufficiency in adults](#)".)

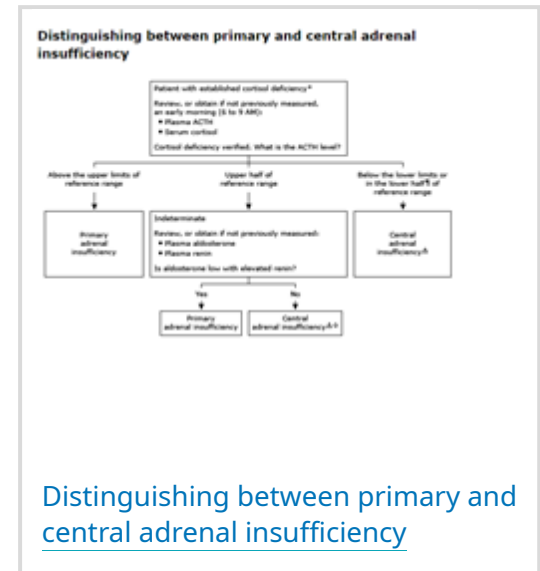
This topic will review the diagnostic steps needed to establish the specific etiology of adrenal insufficiency. The presentation of adrenal insufficiency and the diagnosis of cortisol deficiency reviewed in detail elsewhere. (See "[Clinical manifestations of adrenal insufficiency in adults](#)" and "[Diagnosis of adrenal insufficiency in adults](#)".)

ESTABLISH THE LEVEL OF DEFECT

Once cortisol deficiency is established, we next determine if the defect localizes to the adrenal gland (primary adrenal insufficiency) or the pituitary gland or hypothalamus (secondary or tertiary adrenal insufficiency, respectively). The diagnosis of cortisol deficiency, including basal

and stimulated [cortisol](#) testing, is reviewed separately. (See "[Diagnosis of adrenal insufficiency in adults](#)", section on '[Basal serum cortisol testing](#)' and "[Diagnosis of adrenal insufficiency in adults](#)", section on '[ACTH stimulation tests](#)'.)

ACTH measurement — The plasma [corticotropin](#) (ACTH) level can help distinguish between primary and central (ie, secondary or tertiary) [adrenal insufficiency](#) ([algorithm 1](#)). It can be measured with an early morning serum [cortisol](#) (ie, 6 to 9 AM) as part of the initial evaluation for [adrenal insufficiency](#). Alternatively, ACTH can be measured later in the evaluation after [cortisol](#) deficiency is confirmed; in this case, we remeasure an early morning [cortisol](#) level with the ACTH level to verify [cortisol](#) deficiency and help interpret the ACTH value [1]. (See "[Diagnosis of adrenal insufficiency in adults](#)", section on '[Basal serum cortisol testing](#)'.)



Algorithm 1 - larger image below

Proper specimen collection and handling are critical for ACTH measurement. (See "[Laboratory assessment of hypothalamic-pituitary-adrenal axis function](#)", section on '[Caveats for interpretation](#)'.)

- **ACTH measurement prior to glucocorticoid therapy (optimal)** – Ideally, blood samples for ACTH measurement are drawn before initiating [glucocorticoid](#) therapy because exogenous [glucocorticoids](#) can suppress ACTH secretion. When establishing the cause of [adrenal insufficiency](#) is not urgent (eg, patients with chronic, mild signs and symptoms), treatment decisions may be delayed until diagnostic testing is complete. However, in the setting of moderate to severe signs and symptoms of [adrenal insufficiency](#) or possible [adrenal crisis](#), diagnostic testing should never delay treatment initiation. In such patients, obtaining both an ACTH and [cortisol](#) level as part of the initial evaluation is useful. (See "[Clinical manifestations of adrenal insufficiency in adults](#)", section on '[Adrenal crisis](#)' and "[Treatment of adrenal insufficiency in adults](#)", section on '[Adrenal crisis](#)'.)

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If glucocorticoid therapy has already been started at a physiologic replacement dose – In patients already receiving physiologic [glucocorticoid](#) therapy, blood samples should be drawn prior to the morning dose of a short-acting (eg, [hydrocortisone](#)) or longer-acting (eg, [prednisone](#)) [glucocorticoid](#), assuming that short-acting agents are taken two or three times daily and longer-acting agents are taken once daily in the morning.

- **If glucocorticoid therapy has already been started at a supraphysiologic dose** – High glucocorticoid doses (eg, initiated for possible adrenal crisis) suppress ACTH levels acutely. If supraphysiologic glucocorticoid therapy is given for more than two weeks, ACTH may remain suppressed after the glucocorticoid is discontinued, possibly leading to misdiagnosis of endogenous central adrenal insufficiency. In this setting, ACTH and cortisol levels should be measured at least 24 hours after the last dose of glucocorticoid as an initial screening test; retesting is often required if cortisol is suppressed, particularly after exposure to supraphysiologic glucocorticoid doses for at least six weeks.

ACTH interpretation — If the sample is drawn in the proper setting and the corticotropin (ACTH) assay is reliable, then the plasma ACTH measurement by itself is usually sufficient to discern whether the adrenal insufficiency is primary or central ([algorithm 1](#)) [2,3]. For most two-site chemiluminescent assays, the reference range for ACTH values at 8 AM is between approximately 10 and 50 to 65 pg/mL (2 and 11 to 14 pmol/L). The reference range for the ACTH assay used must be reviewed to interpret the results, since the upper limit varies across assays.

- **ACTH above the reference range** – In patients with confirmed cortisol deficiency, a plasma ACTH concentration above the upper limit of the reference range indicates primary adrenal insufficiency. In patients with primary adrenal insufficiency, cortisol production is deficient, and the loss of cortisol-mediated negative feedback inhibition leads to increased ACTH secretion. In primary adrenal insufficiency, the early morning plasma ACTH concentration is high, usually >2-fold the upper limit of the reference range and sometimes reaching or even exceeding 4000 pg/mL (880 pmol/L) [1]. If the ACTH level is elevated, the next step is to determine the underlying cause of primary adrenal insufficiency. (See 'Evaluation for the cause of primary adrenal insufficiency' below.)
- **ACTH below or in the lower half of the reference range** – In patients with confirmed cortisol deficiency, a low (or inappropriately normal) plasma ACTH concentration (eg, less than approximately 25 to 32 pg/mL [5.5 to 7 pmol/L] in most assays) indicates central (ie, secondary or tertiary) adrenal insufficiency. Patients with central adrenal insufficiency have intrinsically normal adrenal glands that fail to produce cortisol because pituitary ACTH secretion is deficient. Although the ACTH concentration may not be overtly low in central adrenal insufficiency, it is usually in the lower half of the reference range and inappropriately low in the setting of cortisol deficiency. If the ACTH level indicates central adrenal insufficiency, the next step is to determine the underlying etiology. (See 'Evaluation for the cause of central adrenal insufficiency' below.)
- **ACTH in the upper half of the reference range** – In patients with confirmed cortisol deficiency, an ACTH value in the upper half of the reference range (eg, between approximately

25 to 50 pg/mL [5.5 to 11 pmol/L] or 32 to 65 pg/mL [7 to 14 pmol/L] depending on the assay) is considered indeterminate for establishing the level of the defect. An indeterminate level may reflect either evolving primary adrenal insufficiency or partial central adrenal insufficiency. In individuals with cortisol deficiency and recent exposure to supraphysiologic glucocorticoid therapy, an ACTH in the upper half of the reference range or higher likely reflects primary adrenal insufficiency.

In patients in whom the ACTH level is indeterminate, measurement of plasma aldosterone and renin levels and serum electrolytes may help distinguish between primary and central adrenal insufficiency.

- **Low aldosterone with elevated renin** – A low plasma aldosterone level with an elevated renin level is consistent with primary adrenal insufficiency, which results in mineralocorticoid as well as cortisol deficiency. In the early evolution of primary adrenal insufficiency due to autoimmune adrenalitis, mineralocorticoid deficiency may be more pronounced than cortisol deficiency; therefore, a low aldosterone and elevated renin level can precede overt elevation in ACTH [1]. In patients with primary adrenal insufficiency, aldosterone levels are usually low, often in association with hyperkalemia, and provide insufficient mineralocorticoid activity to suppress renin production. If mineralocorticoid deficiency is evident, we next evaluate for the underlying cause of primary adrenal insufficiency. (See ["Hyponatremia and hyperkalemia in adrenal insufficiency"](#) and ["Evaluation for the cause of primary adrenal insufficiency"](#) below.)
 - **Normal aldosterone and renin** – Normal aldosterone and renin levels are consistent with central adrenal insufficiency, which leads to cortisol but not mineralocorticoid deficiency. However, aldosterone and renin may be normal in primary adrenal insufficiency if the underlying cause affects zona fasciculata function alone. For example, preserved aldosterone production may be evident in primary adrenal insufficiency due to adrenal hemorrhage, infectious adrenalitis, or metastatic disease. Further, in central adrenal insufficiency, mineralocorticoid deficiency rarely may evolve after very prolonged deficiency of ACTH.
- If renin and aldosterone levels are normal in a patient who has an indeterminate ACTH level and a clinical history suggestive of cortisol deficiency only under physiologic stress, we make a diagnosis of partial central adrenal insufficiency. With partial central adrenal insufficiency, ACTH production is adequate for cortisol secretion in unstressed conditions but fails to increase appropriately during physiologic stress. Therefore, under unstressed conditions, the serum cortisol value may not be overtly low, and the plasma ACTH level is often indeterminate. Partial adrenal insufficiency is uncommon but may occur, for example, in

patients with central adrenal insufficiency due to pituitary surgery, trauma, or opioid use. (See '[Evaluation for the cause of central adrenal insufficiency](#)' below.)

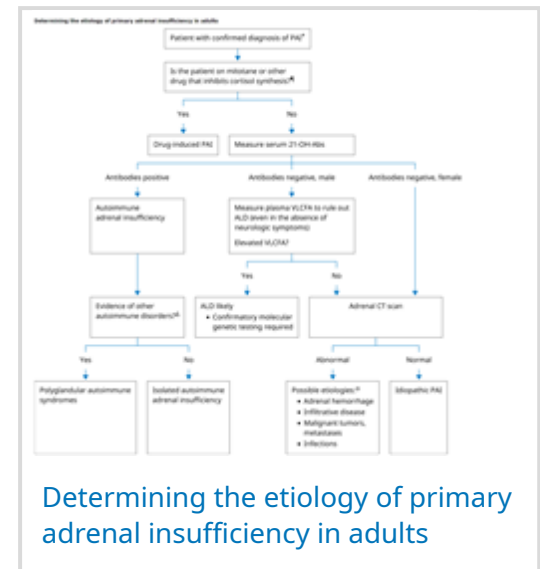
Secondary versus tertiary adrenal insufficiency — Secondary and tertiary adrenal insufficiency can be difficult to differentiate, although this distinction is seldom important from a therapeutic standpoint. In rare cases, this distinction determines the treatment approach, for example, if both exogenous (eg, glucocorticoid or opioid exposure) and endogenous (eg, hypothalamic tumor) causes are considered in the differential etiology. Previously, the ACTH response to administration of corticotropin-releasing hormone (CRH) was used to differentiate between secondary and tertiary adrenal insufficiency. However, as CRH is no longer available, this distinction is typically made on a clinical basis.

EVALUATION FOR THE CAUSE OF PRIMARY ADRENAL INSUFFICIENCY

Once a diagnosis of primary adrenal insufficiency is established, we next determine the underlying etiology ([table 1](#) and [algorithm 2](#)) [1]. (See "[Causes of primary adrenal insufficiency \(Addison disease\)](#)".)

Clinical history

Key elements of history — The patient's age and sex, the clinical setting (eg, anticoagulation or adrenolytic medications), and the presence of other autoimmune endocrine disorders help determine the order of diagnostic tests and, in some cases, may be sufficient to establish the underlying cause [4].



Algorithm 2 - larger image below

Specifically, in patients with primary adrenal insufficiency, we obtain the following components of clinical history:

Relevant symptoms and comorbidities

- **Autoimmunity** – A personal or family history of autoimmune disorders is suggestive of autoimmune adrenal insufficiency. (See "[Causes of primary adrenal insufficiency \(Addison disease\)](#)", section on '[Autoimmune adrenalitis](#)'.)

- **Malignancy** – A personal history of cancer suggests primary adrenal insufficiency due to metastatic disease. Primary malignancies of breast, lung, and colon, as well as melanoma and lymphoma, can metastasize to the adrenal glands. Additionally, primary adrenal lymphoma may cause primary adrenal insufficiency. (See "[Causes of primary adrenal insufficiency \(Addison disease\)](#)", section on 'Metastatic disease'.)
- **Infiltrative disorders** – A personal history of hemochromatosis, sarcoidosis, or primary amyloidosis suggests an infiltrative cause of primary adrenal insufficiency. These infiltrative diseases more commonly cause secondary adrenal insufficiency but rarely can cause primary adrenal insufficiency as well.
- **Anticoagulation or hypercoagulable states** – Heparin or chronic anticoagulation therapy, SARS-CoV-2 infection, or a history of antiphospholipid syndrome suggests the possibility of adrenal hemorrhagic infarction [5-7]. Hypercoagulable states may lead to adrenal infarction due to either hemorrhage or adrenal vein thrombosis. (See "[Causes of primary adrenal insufficiency \(Addison disease\)](#)", section on 'Hemorrhagic infarction'.)
- **Immunocompromise** – Immunocompromise and other risk factors for tuberculosis, human immunodeficiency virus (HIV), and disseminated fungal infections suggest an infectious etiology of primary adrenal insufficiency. Bacteremic colonization of the adrenal glands can lead to adrenal destruction. (See "[Causes of primary adrenal insufficiency \(Addison disease\)](#)", section on 'Infectious adrenalitis'.)
- **Neurologic symptoms** – In males only, a personal or family history of neurologic abnormalities suggests X-linked adrenoleukodystrophy. (See "[Management and prognosis of X-linked adrenoleukodystrophy](#)", section on 'Adrenal insufficiency'.)
- **Medications**
 - **Adrenolytic medications** – Exposure to adrenolytic drugs (eg, [ketoconazole](#), [fluconazole](#), [suramin](#), [mitotane](#), [metyrapone](#), or [etomidate](#)) is consistent with drug-induced adrenal insufficiency. If a patient develops adrenal insufficiency during treatment with an adrenolytic medication and has no risk factors for alternative etiologies, no further diagnostic testing is needed. Medications that may lead to adrenal dysfunction are reviewed in detail elsewhere. (See "[Causes of primary adrenal insufficiency \(Addison disease\)](#)", section on 'Adverse effect of medications'.)
 - **Immune checkpoint inhibitors** – Rarely, exposure to immune checkpoint inhibitors can cause autoimmune primary adrenal insufficiency. Given the rarity of checkpoint inhibitor therapy-induced primary adrenal insufficiency, we exclude other causes prior to assigning

this etiology. (See "[Causes of primary adrenal insufficiency \(Addison disease\)](#)", section on '[Adverse effect of medications](#)'.)

- **Medications that induce cortisol metabolism** – Initiation of growth hormone or thyroid hormone replacement can precipitate acute adrenal insufficiency due to accelerated cortisol metabolism in individuals with underlying dysfunction of the hypothalamic-pituitary-adrenal axis. Therefore, these medications do not constitute causes of adrenal insufficiency but may contribute to its clinical manifestation.
- **History of surgery or trauma** – History of major surgery or physical trauma may implicate bilateral adrenal injury as the cause of primary adrenal insufficiency [8]. In these settings, imaging can be performed to assess for adrenal hemorrhage or other radiographic evidence of injury [9]. (See '[Additional evaluation based on adrenal imaging](#)' below and "[Causes of primary adrenal insufficiency \(Addison disease\)](#)", section on '[Bilateral adrenal injury](#)'.)

Diagnoses made on clinical history alone — In some circumstances, clinical history alone can determine the cause of primary adrenal insufficiency. For example, if no other risk factors for adrenal insufficiency are apparent from clinical history, we make an empiric diagnosis of drug-induced adrenal insufficiency in patients who develop adrenal insufficiency during treatment with an adrenolytic drug such as [ketoconazole](#), [mitotane](#), or [etomidate](#).

Diagnostic testing — If the diagnosis is not apparent from the clinical history, the order of diagnostic tests depends on whether a specific underlying etiology of adrenal insufficiency is suspected. For most people, diagnostic testing should begin with measurement of antibodies to 21-hydroxylase to assess for autoimmune adrenal insufficiency, the most common cause of primary adrenal insufficiency in resource-abundant settings ([algorithm 2](#)) [3]. However, in some cases when an alternative etiology is suspected based on the clinical history (eg, infectious adrenalitis in a patient with known tuberculosis, or metastatic infiltration in a patient with known malignancy), appropriate testing for these etiologies is reasonable prior to measuring anti-21-hydroxylase antibodies.

Measurement of antibodies to 21-hydroxylase — For most patients, we begin with measurement of antibodies to the adrenal enzyme 21-hydroxylase to test for autoimmune adrenalitis. Positive antibody titers for 21-hydroxylase are sensitive (approximately 90 percent) and highly specific (>99 percent) for autoimmune adrenalitis, and antibody titers remain positive at least 30 years after disease onset ([algorithm 2](#)) [10]. (See "[Causes of primary adrenal insufficiency \(Addison disease\)](#)", section on '[Autoimmune adrenalitis](#)'.)

Positive antibodies to 21-hydroxylase — Positive antibodies to 21-hydroxylase confirm an etiology of autoimmune adrenal insufficiency. Patients with autoimmune primary adrenal insufficiency should undergo evaluation for the presence of associated autoimmune endocrinopathies, as may be evident in polyglandular autoimmune syndromes.

- For all patients, we send additional testing for serum calcium, phosphorus, fasting glucose, free thyroxine (T4), and thyroid-stimulating hormone (TSH) levels to evaluate for concurrent autoimmune hypoparathyroidism, diabetes, and hypothyroidism, respectively. If hypocalcemia is found, the serum parathyroid hormone level should be measured. (See ["Causes of primary adrenal insufficiency \(Addison disease\)", section on 'Polyglandular autoimmune syndromes'.](#))
- For females aged <48 years with amenorrhea or oligomenorrhea, we additionally measure serum follicle-stimulating hormone, luteinizing hormone, and estradiol levels. (See ["Evaluation and management of secondary amenorrhea", section on 'Additional evaluation'.](#))
- For males with signs or symptoms of hypogonadism, we also measure serum luteinizing hormone and testosterone levels. (See ["Clinical features and diagnosis of male hypogonadism", section on 'Clinical features'.](#))

Negative antibodies to 21-hydroxylase — If antibody titers against 21-hydroxylase are negative, autoimmune adrenal insufficiency is unlikely. The next diagnostic step to determine the cause of primary adrenal insufficiency depends on patient sex:

- For males only, we next measure plasma levels of very long chain fatty acids (VLCFAs) to assess for adrenoleukodystrophy. We perform this test in males with primary adrenal insufficiency even in the absence of neurologic symptoms because approximately 10 percent of individuals with X-linked adrenoleukodystrophy (ALD) present with idiopathic adrenal insufficiency without accompanying neurologic signs or symptoms. ALD presents in males between two years of age and adulthood but usually before age 7.5 years. Elevated plasma levels of VLCFAs are consistent with ALD, which must be verified with subsequent genetic testing. ALD is reviewed in detail elsewhere. (See ["Clinical features, evaluation, and diagnosis of X-linked adrenoleukodystrophy".](#))
- For females and for males with normal plasma levels of VLCFAs, we next obtain an adrenal protocol, contrast-enhanced computed tomography (CT) scan of the abdomen/pelvis. (See ["Additional evaluation based on adrenal imaging"](#) below.)

Additional evaluation based on adrenal imaging — In most patients with negative anti-21-hydroxylase antibodies, we obtain a contrast-enhanced CT scan of the abdomen/pelvis. Abdominal CT can detect adrenal gland enlargement, areas of necrosis, or adrenal calcification and, accordingly, may suggest an infectious, infiltrative, hemorrhagic, or metastatic cause of primary adrenal insufficiency [11-16].

Abnormal findings on CT often indicate the need for additional evaluation, as follows:

- **Bilateral adrenal gland enlargement** – Bilateral adrenal gland enlargement can suggest an infiltrative process such as infectious adrenalitis or acute adrenal infarction. We pursue additional evaluation guided by clinical history and additional imaging findings.
- **Immunocompromise or known systemic infection** – Tuberculous adrenal insufficiency is the most common infectious cause of primary adrenal insufficiency in non-resource-abundant countries, and patients almost always have obvious active tuberculosis elsewhere [11,17,18]. Additional imaging features of tuberculous adrenalitis include areas of calcification and/or necrosis and peripheral rim enhancement, and atrophic adrenal glands may be evident at advanced stages of infection. Evaluation of extrapulmonary tuberculosis is discussed in detail elsewhere. (See "[Miliary tuberculosis](#)".)

If a particular causative pathogen is not apparent from clinical history, we evaluate for risks of immunocompromise, including testing for HIV ([algorithm 3](#)) in all patients with potential infectious adrenal insufficiency, as this broadens the possible infectious etiologies. As an example, autopsy data suggest that cytomegalovirus is the most common cause of infectious adrenalitis in individuals with HIV infection [19]. Microbiologic testing for other potential causes of infectious adrenalitis, such as *Histoplasma capsulatum*, depends on relevant risk factors. Diagnosis of relevant pathogens are discussed in the respective, dedicated topic reviews. (See "[Diagnosis and treatment of disseminated histoplasmosis in patients without HIV](#)", section on 'Diagnosis of disseminated histoplasmosis' and "[Clinical manifestations and diagnosis of *Cryptococcus neoformans* meningoencephalitis in patients without HIV](#)", section on 'Diagnosis' and "[Approach to the diagnosis of cytomegalovirus infection](#)".)

If history and noninvasive testing do not indicate a particular etiology, CT-guided biopsy of the adrenal glands (for histology; routine, fungal, and mycobacterial cultures; and potentially molecular testing) may be necessary to identify a causative pathogen [20]; however, excluding pheochromocytoma is prudent prior to this procedure, particularly in the setting of adrenal gland enlargement or if the adrenal glands measure >10 Hounsfield units (HU) [18,21]. (See "[Clinical presentation and diagnosis of pheochromocytoma](#)", section on 'Initial biochemical tests'.)

- **Bleeding risk or hypercoagulable state** – Acute adrenal infarction characteristically appears as adrenal enlargement with low attenuation on abdominal CT. However, if infarction occurs in the absence of hemorrhage, attenuation may be normal. Adrenal infarction may result from hemorrhage or adrenal vein thrombosis and may occur in the setting of sepsis, heparin therapy, anticoagulation, hypercoagulable states, or postoperative stress. Measurement of antiphospholipid antibodies can be helpful to diagnose antiphospholipid syndrome in patients who are not taking anticoagulants and have no clear underlying cause of adrenal infarction [22]. The causes of adrenal infarction are discussed in detail elsewhere. (See "[Causes of primary adrenal insufficiency \(Addison disease\)](#)", section on '[Hemorrhagic infarction](#)' and "[Antiphospholipid syndrome: Diagnosis](#)".)
- **Bilateral, focal masses with high attenuation** – Bilateral, well-circumscribed masses with high attenuation (>10 HU) are consistent with adrenal gland metastases. For patients with CT evidence of adrenal gland metastasis and no known primary malignancy, fluorodeoxyglucose (FDG)-positron emission tomography (PET) scan helps verify metastatic involvement of the adrenal glands and localize the primary tumor. When noninvasive testing fails to differentiate between malignant and nonmalignant causes of adrenal insufficiency and this distinction is essential for treatment decisions, we perform ultrasound- or CT-guided fine-needle aspiration (FNA) biopsy of the adrenal glands [20,23,24]. However, pheochromocytoma must be excluded prior to FNA biopsy [21]. The evaluation and imaging characteristics of adrenal metastases and the diagnosis of pheochromocytoma are discussed in detail elsewhere. (See "[Evaluation and management of the adrenal incidentaloma](#)", section on '[Adrenal metastases](#)' and "[Evaluation and management of the adrenal incidentaloma](#)", section on '[Fine-needle aspiration biopsy](#)' and "[Clinical presentation and diagnosis of pheochromocytoma](#)", section on '[Initial biochemical tests](#)'.)

All diagnostic tests are negative — Rarely, patients with primary adrenal insufficiency have negative anti-21-hydroxylase antibodies and normal CT imaging ([algorithm 2](#)).

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If such patients have any risk factors for tuberculous adrenalitis (eg, immunocompromise, known exposure), we pursue additional testing for tuberculosis as abdominal CT findings occasionally can appear normal with tuberculous adrenal insufficiency. (See "[Miliary tuberculosis](#)".)

If this additional evaluation remains unrevealing or the patient has no risk factors for tuberculosis, we presumptively make an etiologic diagnosis based on the likelihood of autoimmunity:

- In patients with a personal history of autoimmunity, we make a presumptive diagnosis of antibody-negative autoimmune primary adrenal insufficiency. Although anti-21-hydroxylase antibody testing is sensitive for diagnosing autoimmune adrenal insufficiency, the sensitivity of this test is not 100 percent.
- In patients without a personal history of autoimmunity, we assign an etiology of idiopathic adrenal insufficiency.

Very rarely, clinical history may suggest a childhood onset of primary adrenal insufficiency due to genetic causes including congenital adrenal hyperplasia (CAH), congenital lipid adrenal hyperplasia, or ACTH insensitivity syndromes. Importantly, these disorders are almost uniformly diagnosed in childhood and not usually considered in the differential etiology for adults with a new diagnosis of primary adrenal insufficiency. (See "[Primary adrenal insufficiency in children: Causes](#)".)

EVALUATION FOR THE CAUSE OF CENTRAL ADRENAL INSUFFICIENCY

Prolonged administration of synthetic glucocorticoids at pharmacologic doses is by far the most common cause of central adrenal insufficiency. Exposure to exogenous glucocorticoids suppresses the secretion of both corticotropin (ACTH) by the pituitary gland (secondary adrenal insufficiency) and corticotropin-releasing hormone (CRH) from the hypothalamus (tertiary adrenal insufficiency).

Common causes of central adrenal insufficiency due to primary pituitary dysfunction are pituitary adenomas, pituitary surgery, or radiation therapy. Other causes of pituitary or hypothalamic disease leading to central adrenal insufficiency include primary tumors (eg, craniopharyngioma) or metastatic disease, autoimmunity, infectious and infiltrative disorders, hypophysitis (medication induced or primary), infarction, and injury due to surgery or trauma. Clinical history alone may be sufficient for a diagnosis of exogenous glucocorticoid-induced adrenal insufficiency. For all other patients, we pursue additional testing and evaluate for the presence of concurrent pituitary hormone deficiencies. (See "[Causes of hypopituitarism](#)" and "[Diagnostic testing for hypopituitarism](#)".)



History of exogenous glucocorticoid use — Adrenal insufficiency due to exogenous glucocorticoid treatment should be suspected in any patient with a recent exposure to glucocorticoid therapy (discontinued within the past three months), particularly in the setting of supraphysiologic doses (≥ 5 mg [prednisone](#) daily or equivalent) administered for ≥ 2 weeks. If adrenal insufficiency is confirmed, a history of recent exposure to supraphysiologic, exogenous glucocorticoid therapy is sufficient to establish glucocorticoid therapy as the cause of adrenal

insufficiency. However, patients who have other risk factors for pituitary disorders or functional suppression (eg, history of autoimmunity, head trauma, history of pituitary tumor, use of opioids or checkpoint inhibitor therapy) may have central adrenal insufficiency due to another cause. In such patients, we perform additional diagnostic testing, including imaging and assessment of other pituitary hormones. (See ['Evaluation for other causes'](#) below and ["Causes of secondary and tertiary adrenal insufficiency in adults"](#), section on ['Chronic high-dose glucocorticoid therapy'](#).)

Patients with glucocorticoid exposure that was remote (>3 months prior), physiologic (3 to 5 mg [prednisone](#) daily or equivalent), or subphysiologic are less likely to have adrenal insufficiency due to glucocorticoid use. However, some individuals exhibit prolonged suppression of the hypothalamic-pituitary-adrenal axis and may manifest signs and symptoms of adrenal insufficiency up to six months or longer after glucocorticoid discontinuation. Similarly, some individuals may develop adrenal insufficiency on subphysiologic oral doses or ocular, inhaled, intra-articular, parenteral, or topical treatments that confer much lower total glucocorticoid exposure. In such patients, we pursue additional diagnostic testing, including imaging and assessment of other pituitary hormones. (See ['Evaluation for other causes'](#) below.)

Notably, patients with suppression of the hypothalamic-pituitary-adrenal axis due to exogenous glucocorticoid therapy are at risk for concurrent hormone deficiencies, including hypogonadism, hypothyroidism, and growth hormone deficiency. If signs and symptoms of these hormone deficiencies are present, additional evaluation is warranted. (See ["Evaluation and management of secondary amenorrhea"](#) and ["Clinical features and diagnosis of male hypogonadism"](#) and ["Diagnostic testing for hypopituitarism"](#), section on ['Growth hormone'](#) and ["Diagnostic testing for hypopituitarism"](#), section on ['Thyrotropin'](#).)

Evaluation for other causes

Clinical history — Key elements of clinical history include head trauma, sellar tumors, recent pituitary surgery, autoimmunity, and medication exposures ([table 2](#)).

- **Relevant symptoms and comorbidities** – A personal or family history of autoimmunity suggests possible lymphocytic hypophysitis. Tuberculosis, sarcoidosis, Langerhans cell histiocytosis, and other granulomatous diseases can cause infiltrative forms of hypophysitis. (See ["Causes of hypopituitarism"](#), section on ['Hypophysitis'](#).)

Tumors including pituitary adenoma and craniopharyngioma can cause central adrenal insufficiency. Surgical or radiation treatment of these tumors also can lead to adrenal insufficiency. Other neoplasms that may impair hypothalamic or pituitary function include glioma, germinoma, teratoma, and metastatic disease. (See ["Causes of hypopituitarism"](#),

[section on 'Hypothalamic disorders'](#) and ["Causes of hypopituitarism"](#), [section on 'Pituitary disorders'.](#))

Headaches, although nonspecific, may suggest a pituitary or suprasellar tumor or hypophysitis. Visual field defects suggest a pituitary tumor with optic chiasm impingement. The clinical or biochemical presence of other pituitary hormone deficiency (hypothyroidism, hypogonadism, growth hormone deficiency) suggests a destructive pituitary process. (See ["Causes of hypopituitarism"](#), [section on 'Pituitary disorders'.](#))

Polyuria and polydipsia suggest concurrent arginine vasopressin deficiency (formerly called central diabetes insipidus), which may occur with central adrenal insufficiency due to pituitary surgery, hypophysitis, infiltrative disorders, or trauma. (See ["Arginine vasopressin deficiency \(central diabetes insipidus\): Etiology, clinical manifestations, and postdiagnostic evaluation"](#).)

- **Medications**

- **Medications that suppress ACTH production** – Certain medications, including glucocorticoids, megestrol, and opioids, can cause central adrenal insufficiency due to ACTH suppression [25]. In patients on chronic opioid therapy, the prevalence of central adrenal insufficiency may approach approximately 30 percent, with higher opioid doses imparting greater risk [26,27].
- **Immune checkpoint inhibitors** – Exposure to immune checkpoint inhibitors can cause central adrenal insufficiency due to lymphocytic hypophysitis [28]. (See ["Causes of hypopituitarism"](#), [section on 'Complication of immunotherapy'](#) and ["Causes of secondary and tertiary adrenal insufficiency in adults"](#), [section on 'Drugs'.](#))
- **Medications that induce cortisol metabolism** – Initiation of growth hormone or thyroid hormone replacement can precipitate acute adrenal insufficiency due to accelerated cortisol metabolism but only in individuals with underlying dysfunction of the hypothalamic-pituitary-adrenal axis. Therefore, these medications do not constitute causes of adrenal insufficiency.



History of trauma or postpartum hemorrhage – Motor vehicle accidents, falls, participation in some sports (eg, soccer or diving) and other causes of head trauma may lead to ACTH deficiency [29]. Pituitary dysfunction may result either from a single episode of severe head trauma or repeated episodes of milder trauma (eg, recurrent falls). Central adrenal insufficiency due to head trauma usually but not exclusively occurs in conjunction

with other pituitary hormone deficiencies. (See ["Causes of hypopituitarism", section on 'Traumatic brain injury'.](#))

Postpartum hemorrhage with hypotension can lead to ischemic infarction of the pituitary gland or hypothalamic regions known as Sheehan syndrome. (See ["Causes of hypopituitarism", section on 'Pituitary infarction \(Sheehan syndrome\)'.](#))

Assessment of other pituitary hormones — In most patients, we perform biochemical testing to evaluate for hypo- or hypersecretion of other pituitary hormones [30]. This evaluation serves two purposes; first, it helps determine whether concurrent pituitary hormone deficiencies are present and warrant treatment. Second, evaluation of pituitary hormone hypo- or hypersecretion can help determine the underlying etiology of adrenal insufficiency (eg, prolactinoma) or narrow the differential diagnosis (eg, presence of posterior pituitary dysfunction excludes pituitary adenoma as the underlying cause). (See ["Diagnostic testing for hypopituitarism"](#) and ["Causes, presentation, and evaluation of sellar masses", section on 'Hormonal evaluation'.](#))

For patients with a history of polydipsia and polyuria, with or without hypernatremia, we also evaluate for arginine vasopressin deficiency. The absence of hypernatremia does not exclude this diagnosis, as individuals with an intact thirst mechanism and unrestricted water access may maintain a normal plasma sodium level. Arginine vasopressin deficiency may result from hypophysitis, trauma, infiltrative disorders (sarcoid, metastases, lymphoma), or pituitary surgery but excludes pituitary adenoma as the underlying cause of central adrenal insufficiency. The presentation and evaluation of arginine vasopressin deficiency is presented elsewhere. (See ["Evaluation of patients with polyuria", section on 'Evaluation of suspected polyuria'.](#))

MRI of the sella turcica and additional evaluation — In most patients with central adrenal insufficiency, we obtain pituitary imaging with contrast-enhanced magnetic resonance imaging (MRI) of the sella turcica [30]. Exceptions include patients who have acutely undergone pituitary surgery and those taking a medication known to cause ACTH suppression (eg, megestrol, opioids) who have no other evidence of pituitary dysfunction.

MRI of the sella turcica can detect primary tumors of the pituitary gland or hypothalamus, metastatic disease, and both infiltrative and lymphocytic forms of hypophysitis. MRI also can detect pituitary apoplexy, abscess, and secondary forms of empty sella syndrome due to prior surgery, radiation, or infarction. In occasional patients with evidence of hypophysitis or metastatic involvement of the pituitary gland, biopsy may be necessary for diagnosis. In patients with other systemic manifestations of infectious or granulomatous disease or malignancy, the specific etiology of the pituitary or hypothalamic lesion usually can be

inferred. However, in patients without a clear underlying diagnosis, we usually proceed with biopsy to better inform prognosis and treatment selection.

Additional testing may be warranted based on imaging findings. (See ["Causes, presentation, and evaluation of sellar masses"](#), section on 'Evaluation of a sellar mass'.)

All diagnostic tests are negative — In occasional patients, diagnostic testing will not reveal any clear etiology of central adrenal insufficiency. For some of these patients, a remote history of severe or repeated head trauma can be elicited. A detailed history to ascertain exposure to exogenous glucocorticoids (including topical, intra-articular, and inhaled formulations) can sometimes be helpful. Very rarely, adults can present with previously undiagnosed adrenal insufficiency due to a genetic cause of ACTH deficiency. These genetic causes are almost uniformly diagnosed during childhood. If no etiology can be identified, adrenal insufficiency is ascribed to idiopathic ACTH deficiency. (See ["Causes of secondary and tertiary adrenal insufficiency in adults"](#), section on 'Genetic causes' and ["Causes of hypopituitarism"](#), section on 'Genetic diseases'.)

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See ["Society guideline links: Adrenal insufficiency"](#).)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more

histicated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical terminology.

Here are the patient education articles that are relevant to this topic. We encourage you to print or e-mail these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topics (see "[Patient education: Addison disease \(The Basics\)](#)" and "[Patient education: Adrenal crisis \(The Basics\)](#)")
- Beyond the Basics topics (see "[Patient education: Adrenal insufficiency \(Beyond the Basics\)](#)")

SUMMARY AND RECOMMENDATIONS

- **Establish the level of the defect** – After a diagnosis of adrenal insufficiency is established, the corticotropin (ACTH) level helps distinguish between primary and central adrenal insufficiency ([algorithm 1](#)).
- **ACTH measurement** – In patients with confirmed cortisol deficiency, an early morning (6 to 9 AM) plasma ACTH can help distinguish between primary and central adrenal insufficiency. It should be interpreted with a concurrently measured serum cortisol level. These tests should be performed prior to initiation of glucocorticoid replacement therapy if possible. (See '[ACTH measurement](#)' above.)

An elevated ACTH level indicates primary adrenal insufficiency, whereas a low or inappropriately normal value indicates central (ie, secondary or tertiary) adrenal insufficiency. (See '[ACTH interpretation](#)' above.)

An ACTH level in the upper half of the reference range is indeterminate and requires additional evaluation with measurement of serum electrolytes and plasma renin and aldosterone levels. A low plasma aldosterone level with an elevated renin level is consistent with primary adrenal insufficiency, a condition that results in mineralocorticoid as well as cortisol deficiency. In the early evolution of primary adrenal insufficiency due to autoimmune adrenalitis, mineralocorticoid deficiency may be more pronounced than cortisol deficiency and therefore precede overt elevation in ACTH. However, aldosterone and renin may be normal in primary adrenal insufficiency if the underlying cause affects zona fasciculata function alone. In central adrenal insufficiency, renin and aldosterone levels are normal. (See '[ACTH interpretation](#)' above.)



Evaluation of primary adrenal insufficiency

Clinical history – Review key elements of the clinical history, including autoimmunity, patient age and sex, medications, and comorbid conditions. If the diagnosis is not apparent ([table 1](#)), measure antibodies to 21-hydroxylase to assess for autoimmune adrenal insufficiency ([algorithm 2](#)). Patients who develop adrenal insufficiency acutely after treatment with adrenolytic medications often do not need further etiologic testing. (See '[Clinical history](#)' above and '[Diagnoses made on clinical history alone](#)' above.)

- **Diagnostic testing** – For most patients, we first measure antibodies to 21-hydroxylase to assess for autoimmune adrenal insufficiency. Additional diagnostic tests for individuals with negative anti-21-hydroxylase antibody titers include plasma levels of very long chain fatty acids (VLCFAs; males only) and contrast-enhanced CT scan of the abdomen/pelvis ([algorithm 2](#)). For patients with evidence of infectious, hemorrhagic, metastatic, or infiltrative adrenal disease, the underlying cause should be identified and treated appropriately. (See '[Diagnostic testing](#)' above.)
- **Evaluation of central (ie, secondary or tertiary) adrenal insufficiency**
 - **Clinical history** – Review key elements of the clinical history, including history of head trauma, sellar tumors, surgery/radiation, and medication exposures, particularly glucocorticoids or opioids. Exposure to exogenous glucocorticoids is the most common cause of central adrenal insufficiency. (See '[History of exogenous glucocorticoid use](#)' above and '[Evaluation for other causes](#)' above.)
 - **Diagnostic testing** – If the diagnosis is not apparent ([table 2](#)), obtain contrast-enhanced MRI of the pituitary and evaluate for other pituitary hormone deficiencies. Patients with recent exposure (within the past three months) to [prednisone](#) ≥ 5 mg daily or equivalent for ≥ 2 weeks do not require MRI unless other risk factors for pituitary dysfunction are present. For individuals with evidence of hypo- or hypersecretion of other pituitary hormones or those for whom the etiology remains uncertain, we proceed with imaging. Contrast-enhanced MRI of the sella turcica can detect adrenal insufficiency due to lymphocytic, infiltrative, or infectious forms of hypophysitis; sellar tumors; pituitary metastases; or secondary empty sella syndrome. (See '[Assessment of other pituitary hormones](#)' above and '[MRI of the sella turcica and additional evaluation](#)' above.)

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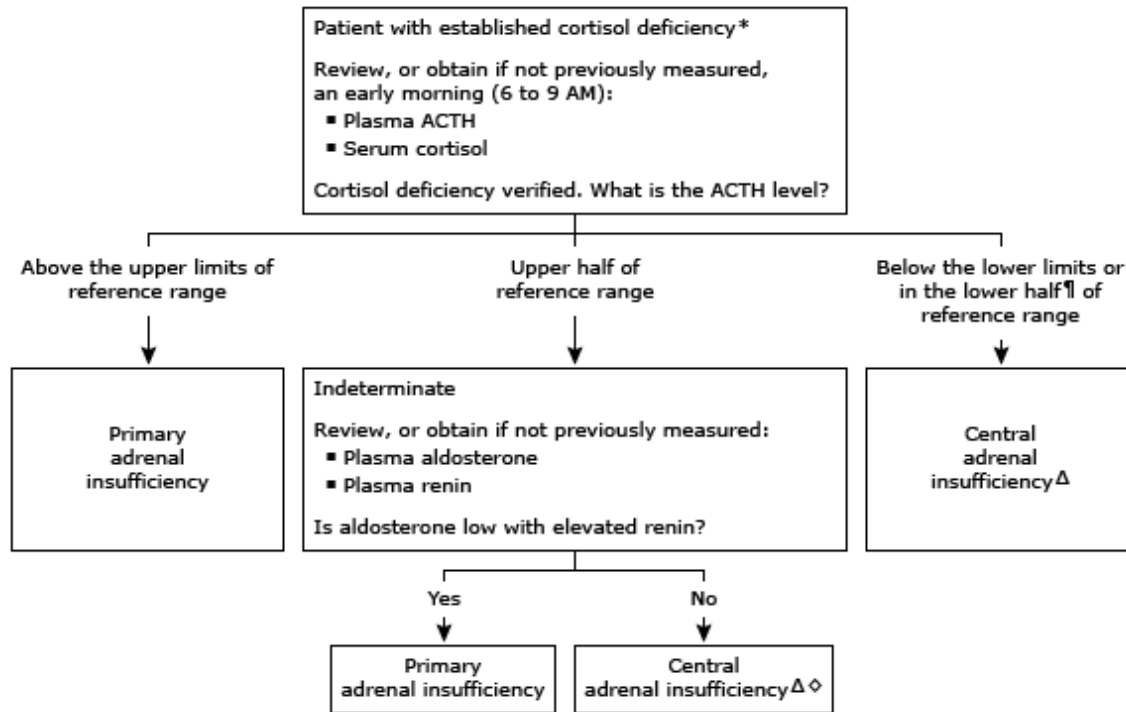
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GRAPHICS

Algorithm 1: Distinguishing between primary and central adrenal insufficiency



Adrenal insufficiency can be caused by diseases of the adrenal gland (primary), interference with ACTH secretion by the pituitary gland (central, also called secondary adrenal insufficiency), or interference with CRH secretion by the hypothalamus (central, also called tertiary adrenal insufficiency).

ACTH: corticotropin; CRH: corticotropin-releasing hormone.

* Refer to UpToDate content for basal and stimulated serum cortisol thresholds that identify cortisol deficiency.

¶ Although the ACTH concentration may not be overtly low in central adrenal insufficiency, it is usually in the lower half of the reference range and inappropriately low in the setting of cortisol deficiency.

Δ The distinction between secondary (interference with ACTH secretion) and tertiary (interference with CRH secretion) adrenal insufficiency is usually made on a clinical basis alone, although this distinction is seldom important from a therapeutic standpoint.

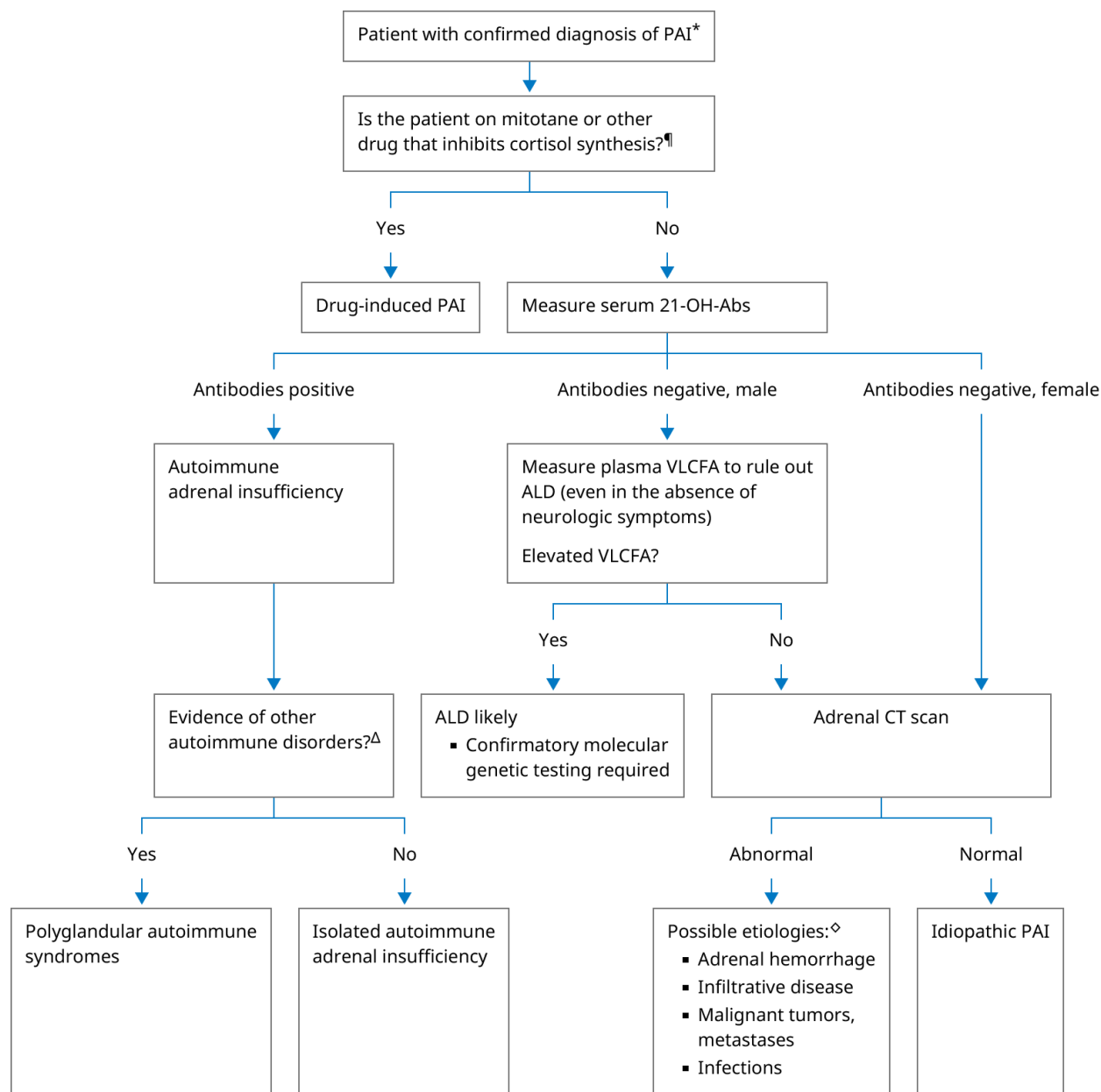
✦
a Plasma levels of renin and aldosterone are usually unaffected in central adrenal insufficiency. If clinical history suggests cortisol deficiency only under physiologic stress, partial central adrenal insufficiency is possible (eg, due to pituitary surgery, opioid use). With partial central adrenal insufficiency, ACTH production is adequate for cortisol secretion in unstressed conditions but fails to increase appropriately during physiologic stress. Under unstressed conditions, the serum cortisol value may not be overtly low and the plasma ACTH level is often indeterminate.

Table 1: Causes of primary adrenal insufficiency

Autoimmune adrenalitis
- Isolated adrenal insufficiency - Polyglandular autoimmune syndrome type 1 - Polyglandular autoimmune syndrome type 2
Infectious adrenalitis
- Tuberculosis - Disseminated fungal infection - Histoplasmosis - Paracoccidioidomycosis - HIV infection and AIDS - Syphilis - African trypanosomiasis
Metastatic cancer
Primarily lung, breast, stomach, and colon cancer, or lymphoma
Adrenal hemorrhage or infarction
Drugs
- Ketoconazole - Fluconazole - Rifampin - Phenytoin - Barbiturates - Other: etomidate, metyrapone, suramin, mitotane, osilodrostat, levoketoconazole, immune checkpoint inhibitors
Other
- Adrenoleukodystrophy and adrenomyeloneuropathy - Congenital adrenal hyperplasia - Familial glucocorticoid deficiency - Familial glucocorticoid resistance - Defective cholesterol metabolism

AIDS: acquired immunodeficiency syndrome; HIV: human immunodeficiency virus.

Algorithm 2: Determining the etiology of primary adrenal insufficiency in adults



This algorithm reviews the approach to determining the etiology of primary adrenal insufficiency. It is based upon UpToDate content on diagnosis of adrenal insufficiency and causes of primary adrenal insufficiency.

☆
a 21-OH-Abs: 21-hydroxylase antibodies; ALD: adrenoleukodystrophy; CT: computed tomography; PAI: primary adrenal insufficiency; VLCFA: very long-chain fatty acids.

* Congenital adrenal hyperplasia due to 21-hydroxylase deficiency is not included in the algorithm, because it presents in infancy or childhood, not adulthood.

¶ The adrenocorticolytic drug mitotane causes PAI. Other drugs that inhibit cortisol biosynthesis (eg, ketoconazole, osilodrostat, metyrapone) do not usually cause clinically important PAI unless the patient has limited pituitary or adrenal reserve.

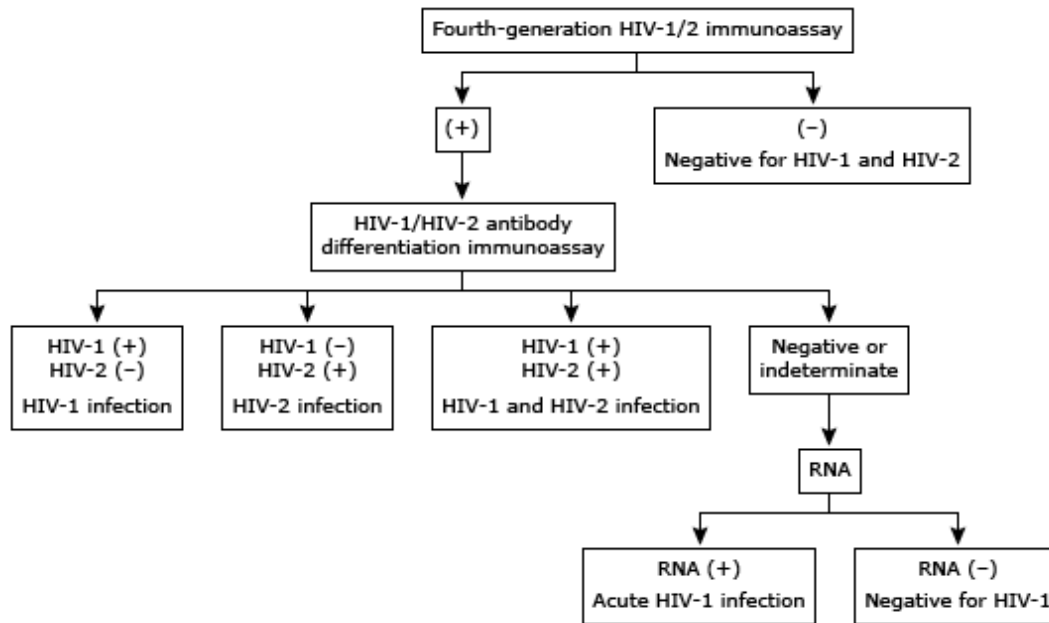
Δ In adults with autoimmune primary adrenal insufficiency, thyroid disease and type 1 diabetes are the most likely endocrine disorders to also be present (as part of polyglandular autoimmune syndrome type II).

◇ To review the characteristic adrenal CT findings of these disorders, refer to UpToDate topic on causes of primary adrenal insufficiency.

Adapted from: Bornstein SR, Allolio B, Arlt W, Barthel A, et al. Diagnosis and treatment of primary adrenal insufficiency: An Endocrine Society clinical practice guideline. J Clin Endocrinol Metab 2016; 101:364

Graphic 113308 Version 3.0

Algorithm 3: Recommended algorithm for HIV diagnosis



HIV: human immunodeficiency virus.

Modified from: CDC and Prevention and Association of Public Health Laboratories. Laboratory Testing for the Diagnosis of HIV Infection: Updated Recommendations. Available at <http://stacks.cdc.gov/view/cdc/23447>. Published June 27, 2014.

Graphic 91270 Version 2.0

Table 2: Major causes of hypopituitarism

Hypothalamic diseases
Mass lesions – Benign (craniopharyngiomas) and malignant tumors (metastatic from lung, breast, etc)
Radiation – For CNS and nasopharyngeal malignancies
Infiltrative lesions – Sarcoidosis, Langerhans cell histiocytosis
Infections – Tuberculous meningitis
Other – Traumatic brain injury, stroke
Pituitary diseases
Mass lesions – Pituitary adenomas, other benign tumors, cysts
Pituitary surgery
Pituitary radiation
Infiltrative lesions – Hypophysitis, hemochromatosis
Infection/abscess
Infarction – Sheehan syndrome
Apoplexy
Genetic mutations
Empty sella
Medications
Glucocorticoids
Opiates
Immune checkpoint inhibitors
High-dose progestogens (eg, megestrol acetate)

CNS: central nervous system.



Graphic 55951 Version 6.0

Contributor Disclosures

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